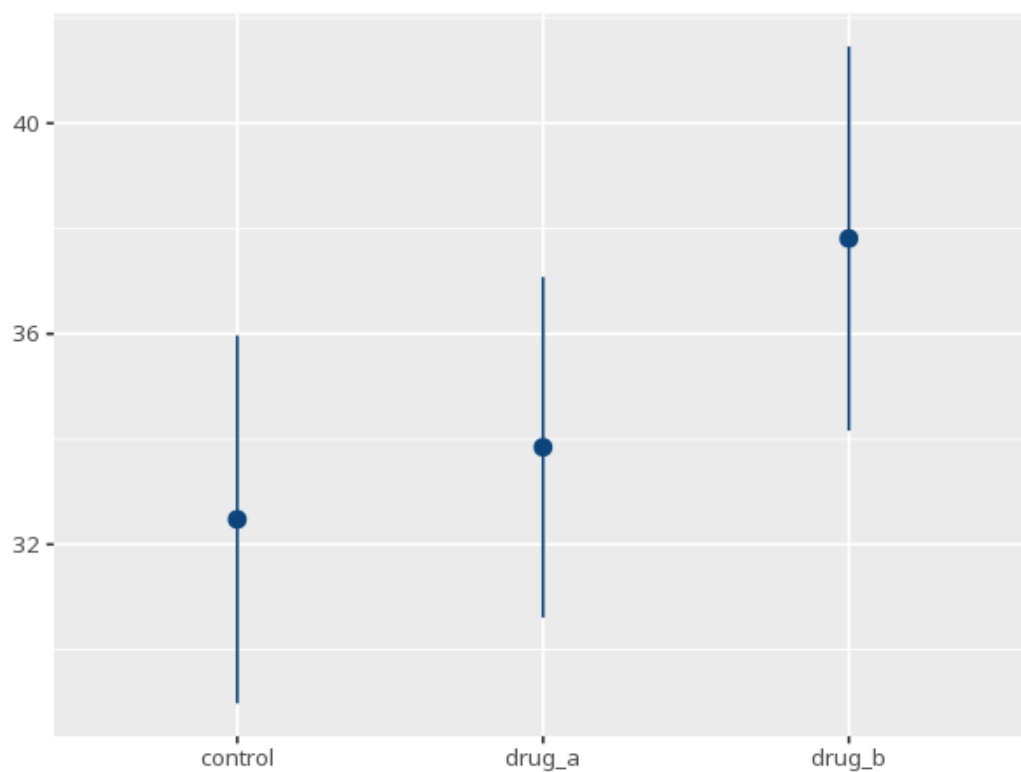


1 One-way ANOVA + post-hoc

Group	N	Mean	Std. Dev.
control	20	32.48	7.47
drug_a	20	33.84	6.91
drug_b	20	37.81	7.79

Source	SS	df	MS	F	p	η^2 [95% CI]
Between	307.07	2	153.53	2.81	.069	0.09 [0.00, 1.00]
Within	3119.95	57	54.74			

Pair	Mdiff	SE	padj	95% CI
control - drug_a	-1.37	2.34	.828	[-7.00, 4.26]
control - drug_b	-5.33	2.34	.067	[-10.96, 0.29]
drug_a - drug_b	-3.96	2.34	.216	[-9.59, 1.66]



The F and its p tell you whether the groups differ overall; η^2 (or ω^2) is the effect size. If significant, the post-hoc table shows which specific pairs differ, with multiplicity-adjusted p-values. Your significance threshold (α) is 0.05.

A one-way ANOVA gave $F(2,57)=2.81$, $p = .069$, $\eta^2=.09$ [.00, 1.00]. Tukey HSD post-hoc tests showed...

Statistical basis: F-test for equality of several means (ANOVA) (Fisher, R. A. (1925). *Statistical Methods for Research Workers*. Oliver & Boyd.); Tukey HSD post-hoc comparison (Tukey, J. W. (1949). "Comparing individual means in the analysis of variance." *Biometrics*, 5(2), 99-114.); Bonferroni-corrected post-hoc comparison (a 'posthoc' choice) (Dunn, O. J. (1961). "Multiple comparisons among means." *Journal of the American Statistical Association*, 56(293), 52-64.); Scheffé-corrected post-hoc comparison (a 'posthoc' choice) (Scheffé, H. (1953). "A method for judging all contrasts in the analysis of variance." *Biometrika*, 40(1-2), 87-104.); Levene's test for equal variances (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), *Contributions to Probability and Statistics* (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." *Journal of the American Statistical Association*, 69(346), 364-367.); η^2 effect-size benchmarks (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.).